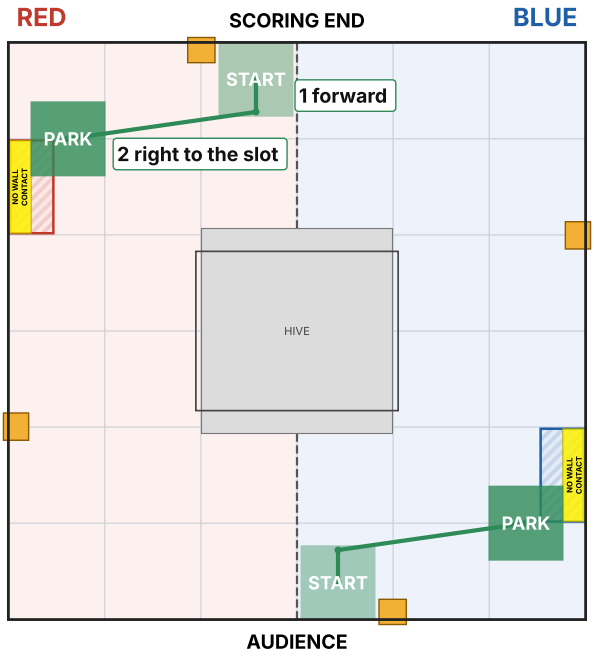
Odo Supporting Shooter Auto

Use this when our partner shoots first. We wait for them to tip the hive, then we shoot.

The robot waits 12 seconds, shoots, then drives to the parking slot on its own. It measures where it is as it drives (odometry).

What the robot does



Time	What happens
0 - 12 s	Waits. Just before 12 s it backs gently into the wall to line itself up. This is normal.
12 - 15 s	Launcher spins up.
15 - 22.5 s	Shoots the pollen.
about 22.5 - 26 s	Drives: (1) forward, clear of the flower, (2) right to the **OUTER** parking slot. Then it stops.

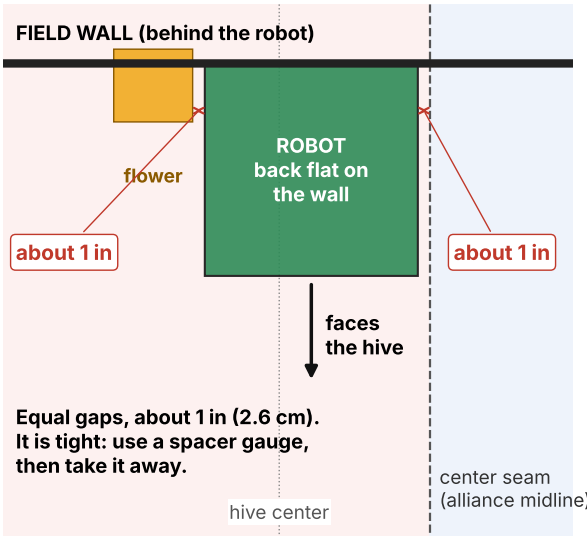
Points if it works: **LEAVE 3 + PARK 5**. Our partner parks in the **INNER** slot.

Time is tight. The robot can't move until it has finished shooting at 22.5 s, and our program stops everything at 27 s to be safe. Don't add delay or feed time unless the coach says.

Before the match

1. Agree with the partner. They shoot first, finish by about **14 s**, then drive away to the INNER slot. We start shooting at **15 s**. Show them the pit sheet.

2. Put the robot on the field.



- **Red:** at the **scoring-end** wall. **Blue:** at the **audience** wall.
- Back of the robot **flat against the wall**, facing the hive.
- **Between the flower and the center seam, with the same gap on both sides: about 1 in (2.6 cm).** It's tight. Use a spacer gauge to set the gaps, then take it away before the match.
- Exactly **4 pollen** loaded. **Not touching the flower.** Not in the loading zone.

3. On the Driver Hub, pick **Odo Supporting Shooter Auto**. It's in the autonomous list, group *Pollen Odometry*.

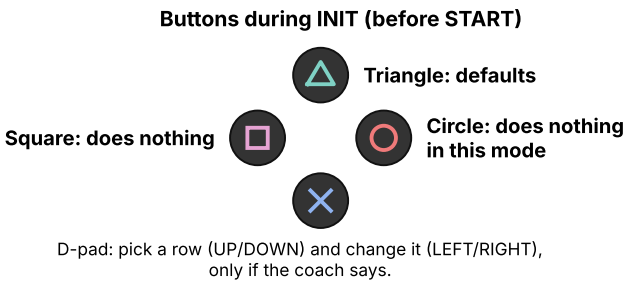
4. Press INIT. Don't touch the robot for 1 second: the odometry sensor is getting ready.

5. Press Triangle once. That puts every setting back to the defaults, including the 12 s wait. Then make only the changes the coach asks for. A different wait is a *Start delay seconds* change, if the partner needs it.

6. Check the screen:

Line	Should say
Build	`pollen-odo-v5`
Mode	`ODOMETRY Supporting Shooter`
Odometry x fwd / y left / heading	numbers near 0 (not `NOT VALID`)
Settings notice	`Defaults: forward clear of flower, then right to the planned park spot`
Start delay seconds	`12.00`
Status	`READY - no movement until START`

7. Press START when the match starts. The robot will sit still for 12 seconds. That's correct.



- **Triangle:** back to the defaults. Always safe.
- **Square** and **Circle:** do nothing in this mode.
- **D-pad:** changes settings. Only if the coach says.

If something is wrong

You see	What it means	Do this
`BLOCKED: ... No goBILDA Pinpoint named "odometry"`	The odometry sensor isn't found	Use **Pollen Supporting Shooter Auto** instead (timed fallback). Tell a programmer.
`Odometry ...: NOT VALID`	The odometry sensor isn't reading	Press STOP, then INIT again with the robot still. Still bad: use the timed fallback.
`READY FOR DEPARTURE ONLY - SHOOTER FAILED`	The launcher didn't start up	The robot still waits out the shooting time, then parks. Tell a programmer.
`NOT READY - over time budget`	Someone's settings take too long	Press Triangle.
Any other `BLOCKED`	Something isn't plugged in, or the wrong robot configuration is active	Tell a programmer.

After the match, tell the programmers

- Did the partner tip the hive before 15 s? Did we shoot all 4? Did we park in the slot?
- Did the robot touch the flower when it drove forward?
- If it stopped before parking: the last **Phase** shown (`TIME LIMIT - STOPPED` means it ran out of time), and anything under **Warning**.